1. Describe the design and constructional features used in current transformer to reduce the errors.
2. Explain the methods of turns compensation used in current transformers to reduce the ratio error. The explanation should be illustrated with the help of a suitable example.
3. A ring-core current transformer with a nominal ratio of 500/5 and a bar primary has a secondary resistance of 0.5Ω and negligible secondary reactance. The resultant of magnetizing and iron loss components of the primary current associated with a full load secondary current of 5 A in a burden of 1.0 Ω (non-inductive) is 3 A at a power factor of 0.4. Calculate the true ratio and the phase angle error of transformer on full load. Calculate also the total flux in the core assuming a frequency of 50 Hz.
4. A current transformer of turn ratio 1:200 is rated as 1000/5A, 25VA The core loss and magnetising component of primary current are 4A and 7.5A under rated conditions. Determine the phase angle and percentage ratio errors for the rated burden and rated secondary current at 0.8 p.f. legging and 0.8 p.f. leading. Neglect the resistance & leakage reactance of secondary winding.
5. A current transformer of nominal ratio 1000/5 A, is operating with total secondary impedance 0.4+j0.3Ω. At rated current the components of the primary current associated with the core-magnetizing and core loss effects are respectively 6 A and 1.5 A. The primary winding has 4 turns. Calculate the ratio error and phase angle at rated primary current if the secondary winding has (a) 800 turns (b) 795 turns.